1.6 Accuracy and Estimation

Name _____ Period ____ Date _____

Algebra I 1.6 · Student sheet

I can…

- estimate results with an appropriate level of accuracy.
- decide how many items I need and round correctly in context.
- choose an appropriate approximation for a measurement.

Vocabulary · fill in the missing word

Estimate — an _____ answer found by rounding

Accuracy — how close a measured or _____ value is to the actual value

Round up in context — when you need _____ (bags, buses, cans), always round up

Part 1 · Garden supply run

Mulch: a 12 ft by 8 ft garden bed, 3 inches deep. Bags hold 2 ft³ or 2.5 ft³.

Volume needed: _____ Bags (2 ft³): _____ Bags (2.5 ft³): _____

Paint: 980 ft² of fence. One gallon covers 350 ft².

Gallons: _____ Gallons to buy: _____

Why can't you round down for these? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

Estimate each result.

a. 19.8×5.1 b. $\sqrt{50}$

Round up: When you need enough of something, always round _____.

Example 2

A sandbox is 6 feet by 5 feet and 1.5 feet deep. Sand comes in bags of 0.8 cubic foot. How many bags do you need?

Accuracy: A measurement can be only as accurate as the _____ used to measure it.

Final answers: Round a final answer to a _____ level of accuracy for the situation.

Example 3

A student says the hallway is 152.387 meters long. Is that an appropriate measurement? What would be better?

Part 3 · Partner practice

Partner 1

Estimate 49.7 ÷ 9.8.

Partner 2

One bag of grass seed covers 15 square feet. How many bags do you need for a 100-square-foot lawn?